Data Mining:

Concepts and Techniques

(3rd ed.)

— Chapter 1 —

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Chapter 1. Introduction

- Why Data Mining?
- What Is Data Mining?
- A Multi-Dimensional View of Data Mining
- What Kind of Data Can Be Mined?
- What Kinds of Patterns Can Be Mined?
- What Technology Are Used?
- What Kind of Applications Are Targeted?
- Major Issues in Data Mining
- A Brief History of Data Mining and Data Mining Society
- Summary

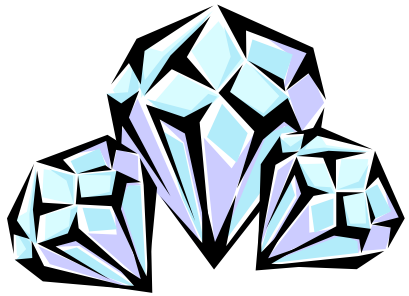
Why Data Mining?

- The Explosive Growth of Data: from terabytes to petabytes
  - Data collection and data availability
    - Automated data collection tools, database systems, Web, computerized society
  - Major sources of abundant data
    - Business: Web, e-commerce, transactions, stocks, …
    - Science: Remote sensing, bioinformatics, scientific simulation, …
    - Society and everyone: news, digital cameras, YouTube
- We are drowning in data, but starving for knowledge!
- "Necessity is the mother of invention"—Data mining—Automated analysis of massive data sets

What Is Data Mining?

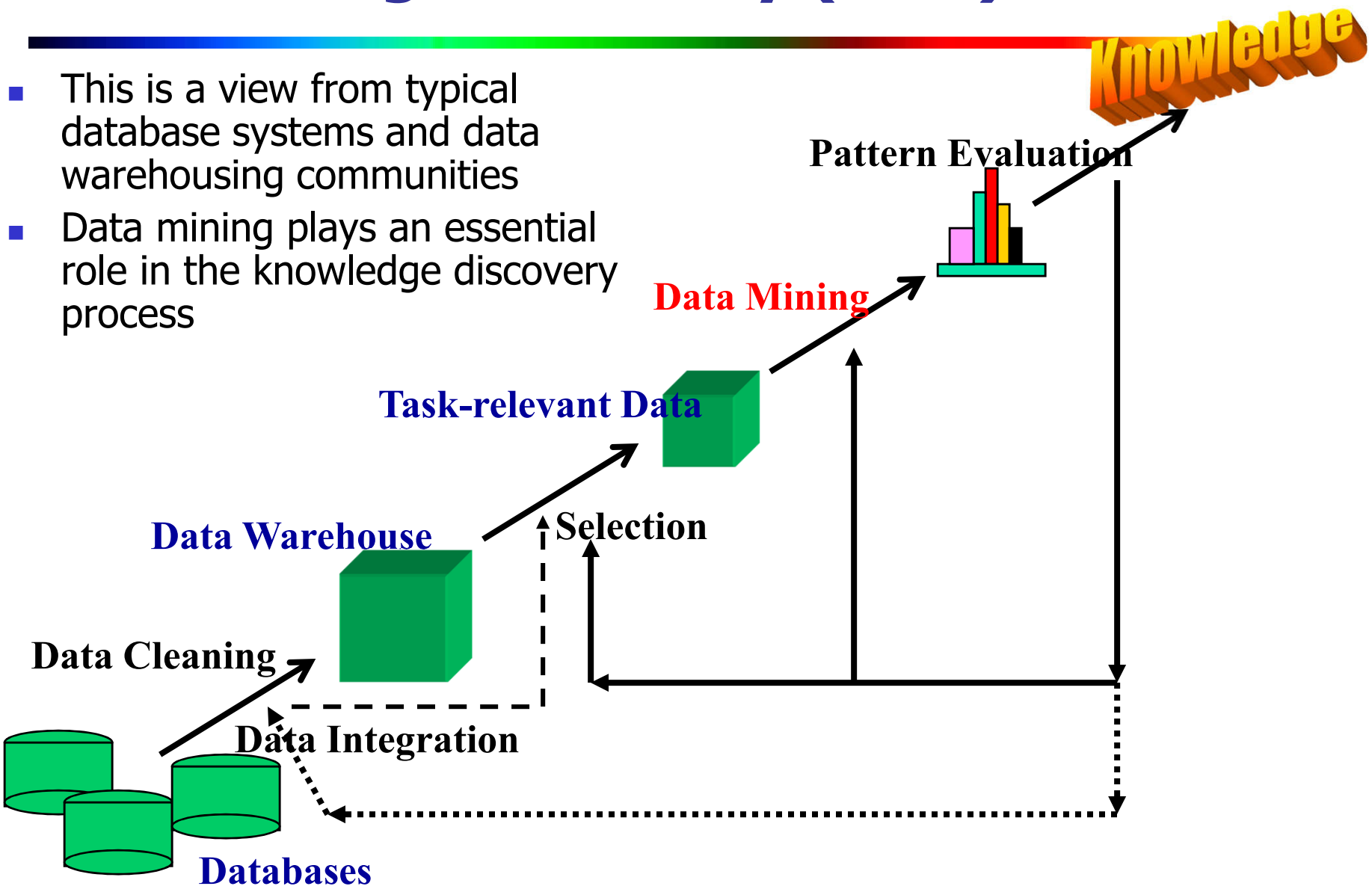


- Data mining (knowledge discovery from data)
  - Extraction of interesting (non-trivial, implicit, previously unknown and potentially useful) patterns or knowledge from huge amount of data
  - Data mining: a misnomer?
- Alternative names
  - Knowledge discovery (mining) in databases (KDD), knowledge extraction, data/pattern analysis, data archeology, data dredging, information harvesting, business intelligence, etc.
- Watch out: Is everything "data mining"?
  - Simple search and query processing
  - (Deductive) expert systems



Knowledge Discovery (KDD) Process

- This is a view from typical database systems and data warehousing communities
- Data mining plays an essential role in the knowledge discovery process



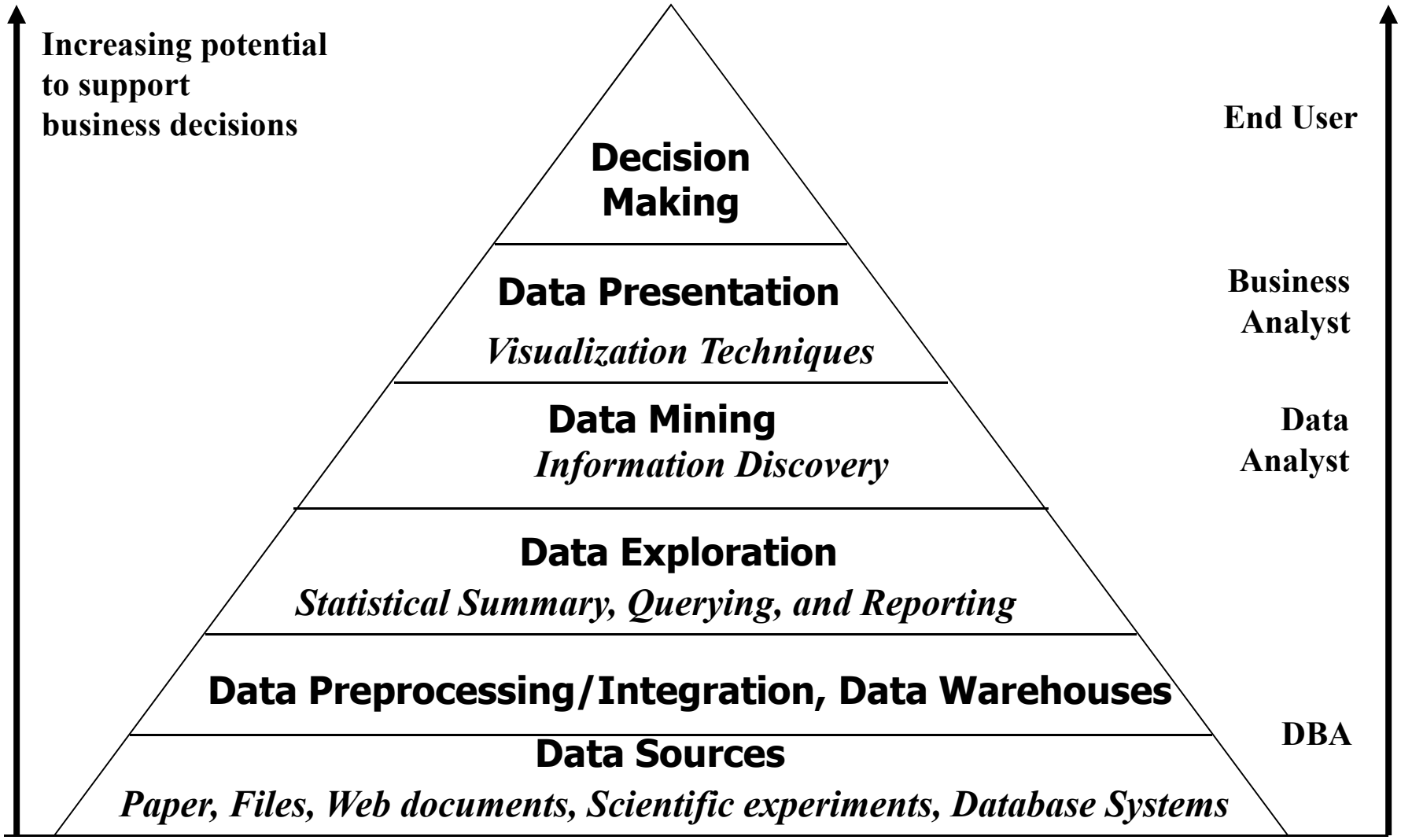
Example: A Web Mining Framework

- Web mining usually involves
  - Data cleaning
  - Data integration from multiple sources
  - Warehousing the data
  - Data cube construction
  - Data selection for data mining
  - Data mining
  - Presentation of the mining results
  - Patterns and knowledge to be used or stored into knowledge-base

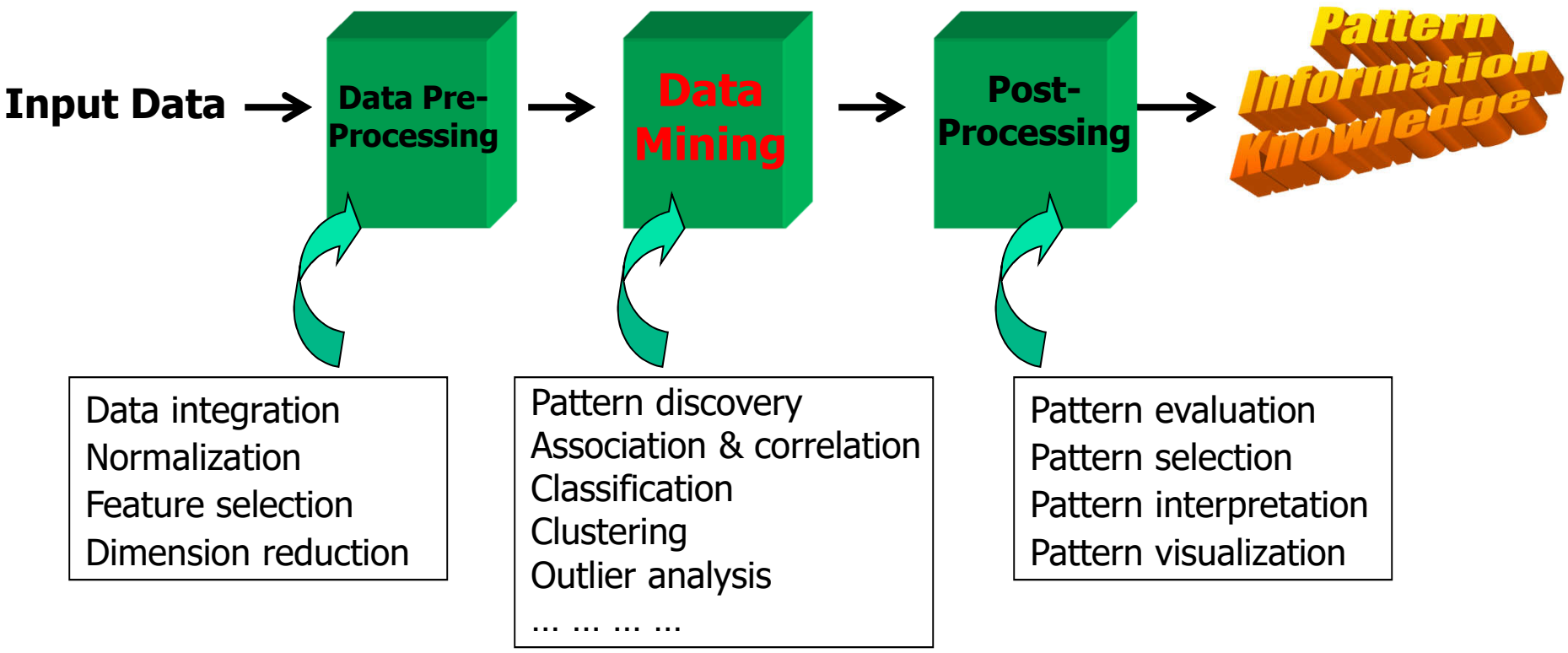
A Web Mining Framework: Data Cube

- A data cube is a multi-dimensional data structure.
- A data cube is characterized by its dimensions (e.g., Products, States, Date).
- Each dimension is associated with corresponding attributes (for example, the attributes of the Products dimensions are T-Shirt, Shirt, Jeans and Jackets)

Data Mining in Business Intelligence



KDD Process: A Typical View from ML and Statistics



- This is a view from typical machine learning and statistics communities

Example: Medical Data Mining

- Health care & medical data mining – often adopted such a view in statistics and machine learning
- Preprocessing of the data (including feature extraction and dimension reduction)
- Classification or/and clustering processes
- Post-processing for presentation

Multi-Dimensional View of Data Mining

- **Data to be mined**
  - Database data (extended-relational, object-oriented, heterogeneous, legacy), data warehouse, transactional data, stream, spatiotemporal, time-series, sequence, text and web, multi-media, graphs & social and information networks
- **Knowledge to be mined (or: Data mining functions)**
  - Characterization, discrimination, association, classification, clustering, trend/deviation, outlier analysis, etc.
  - Descriptive vs. predictive data mining
  - Multiple/integrated functions and mining at multiple levels
- **Techniques utilized**
  - Data-intensive, data warehouse (OLAP), machine learning, statistics, pattern recognition, visualization, high-performance, etc.
- **Applications adapted**
  - Retail, telecommunication, banking, fraud analysis, bio-data mining, stock market analysis, text mining, Web mining, etc.

Data Mining: On What Kinds of Data?

- Database-oriented data sets and applications
  - Relational database, data warehouse, transactional database
- Advanced data sets and advanced applications
  - Data streams and sensor data
  - Time-series data, temporal data, sequence data (incl. bio-sequences)
  - Structure data, graphs, social networks and multi-linked data
  - Object-relational databases
  - Heterogeneous databases and legacy databases
  - Spatial data and spatiotemporal data
  - Multimedia database
  - Text databases
  - The World-Wide Web

Relational Database: Example

- The fictitious AllElectronics is described by the following relation tables:
  - customer, item, employee, and branch
- The relation customer consists of a set of attributes describing the customer information, including a unique customer identity number (cust ID), customer name, address, age, occupation, annual income, credit information, and category.
- Similarly, each of the relations item, employee, and branch consists of a set of attributes describing the properties of these entities.
- Tables can also be used to represent the relationships between or among multiple entities:
  - purchases (customer purchases items, creating a sales transaction handled by an employee)
  - items sold (lists items sold in a given transaction)
  - works at (employee works at a branch of AllElectronics)

AllElectronics database

- Relational schema for the AllElectronics relational database, :

customer	(*cust_ID, name, address, age, occupation, annual_income, credit_information, category, . . .*)
item	(*item_ID, brand, category, type, price, place_made, supplier, cost, . . .*)
employee	(*empl_ID, name, category, group, salary, commission, . . .*)
branch	(*branch_ID, name, address, . . .*)
purchases	(*trans_ID, cust_ID, empl_ID, date, time, method_paid, amount*)
items_sold	(*trans_ID, item_ID, qty*)
works_at	(*empl_ID, branch_ID*)

Query vs. mining

- Relational data can be accessed by database queries written in a relational query language (e.g., SQL) .
- A given query is transformed into a set of relational operations, such as join, selection, and projection. A query allows retrieval of specified subsets of the data.
- Example: "Show me a list of all items that were sold in the last quarter"
- Relational languages also use aggregate functions such as sum, avg, count, max, and min. Using aggregates allows you to ask:
- "Show me the total sales of the last month, grouped by branch,"
- "How many sales transactions occurred in the month of December?"
- "Which salesperson had the highest sales?"
- …

Query vs. mining (cont.)

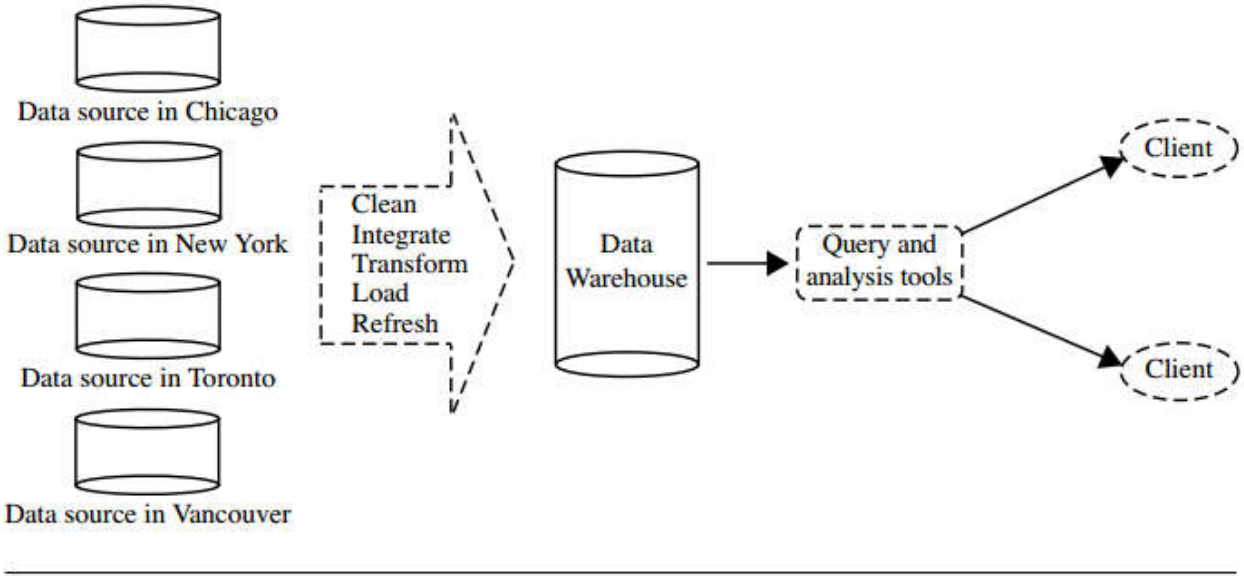
- When **mining** relational databases, we can go further by searching for trends or data patterns.
- For example, data mining systems can analyze customer data to predict the credit risk of new customers based on their income, age, and previous credit information.
- Data mining systems may also detect deviations—that is, items with sales that are far from those expected in comparison with the previous year.
- Such deviations can then be further investigated. For example, data mining may discover that there has been a change in packaging of an item or a significant increase in price.

Data Warehouse

- Suppose that AllElectronics is a successful international company with branches around the world. Each branch has its own set of databases.
- The president of AllElectronics has asked you to provide an analysis of the company's sales per item type per branch for the third quarter. This is a difficult task, particularly since the relevant data are spread out over several databases physically located at numerous sites.
- A data warehouse is a repository of information collected from multiple sources, stored under a unified schema, and usually residing at a single site.
- Data warehouses are constructed via a process of data cleaning, data integration, data transformation, data loading, and periodic data refreshing.

Data Warehouse (cont.)

- To facilitate decision making, the data in a data warehouse are organized around major subjects (e.g., customer, item, supplier, and activity).
- The data are stored to provide information from a historical perspective, such as in the past 6 to 12 months, and are typically **summarized**. For example, rather than storing the details of each sales transaction, the data warehouse may store a summary of the transactions per item type for each store.



Extract, Transform, and Load

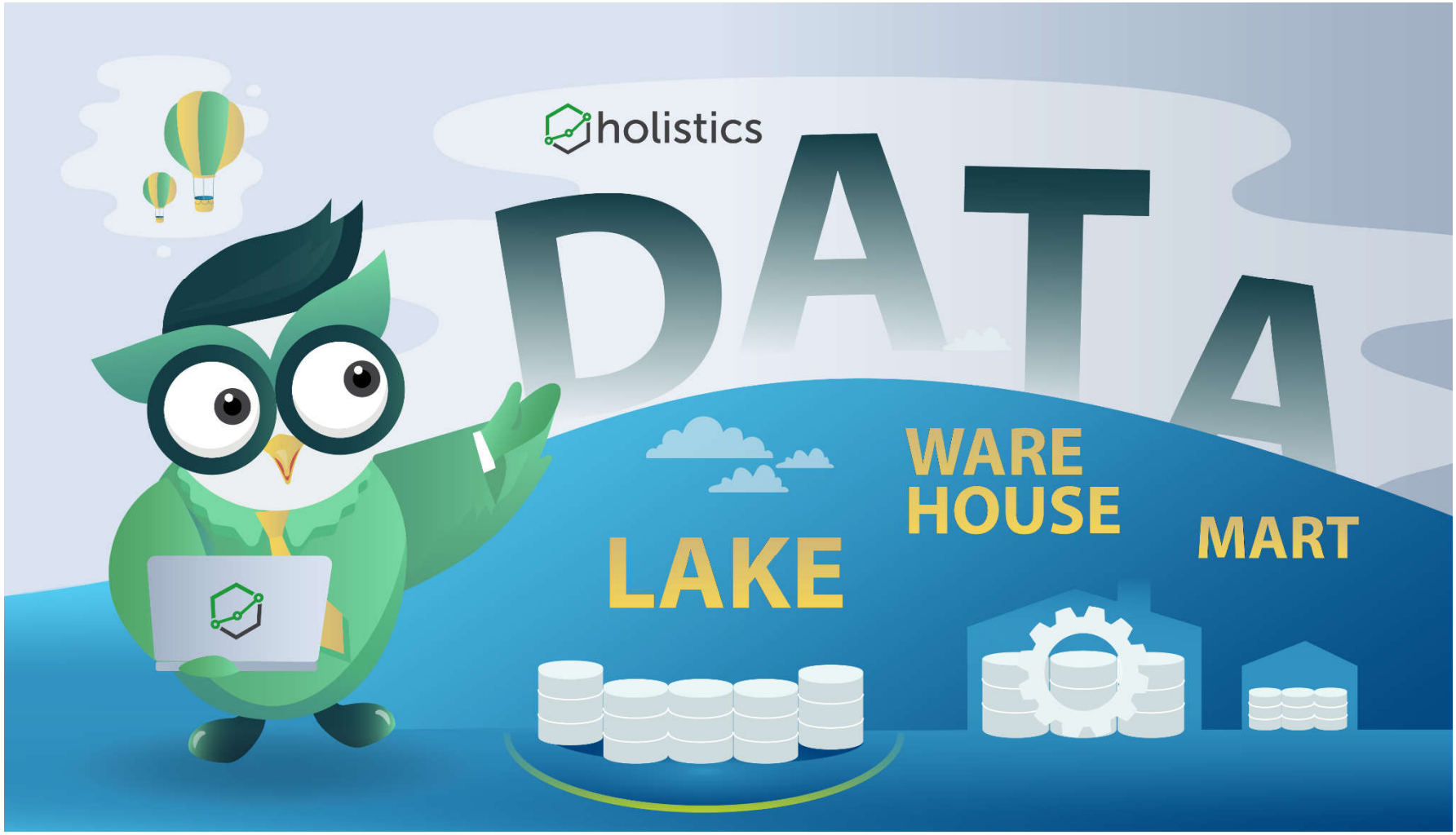
- **ETL** in data mining consists of the construction of new data subsets derived from existing data sources.

- **ETL** stands for the whole process of taking data from various sources and combining it, transforming it, and loading big data using database tools.

- **Extract** is to get data out of different data sources.
- **Transform** means to change the data format in order to better support querying and analysis.
- **Load** is to get this data into a target storage.

- **Data warehouse** is a system that actually performs some ETL operations: extract, clean, conform and deliver source data into a dimensional data store and then support and implement querying and analysis for the purpose of decision making.

Data Lake vs Data Warehouse vs Data Mart



Data Lake

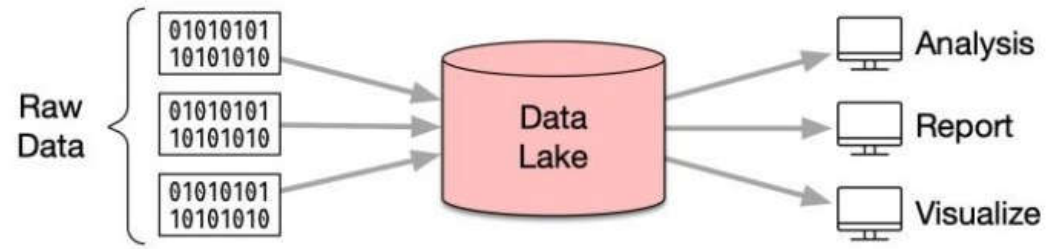
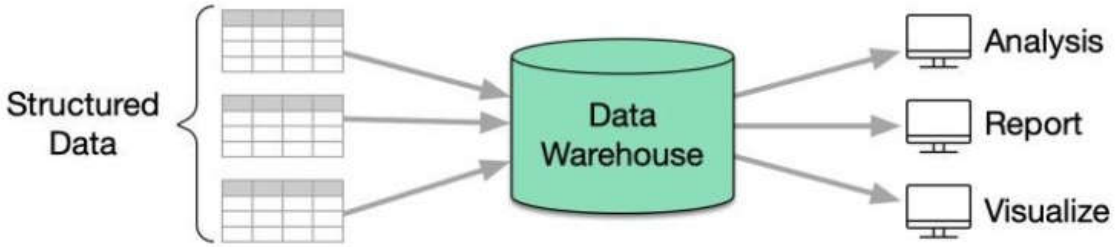
- A data lake is the place where you dump all forms of data generated in various parts of your business: structured data feeds, chat logs, emails, images (of invoices, receipts, checks etc.), and videos.

- The data collection routines does not filter any information out; data related to canceled, returned, and invalidated transactions will also be captured, for instance.

Data Mart

- A data mart is a data storage system that contains information specific to an organization's business unit.
- It contains a small and selected part of the data that the company stores in a larger storage system.

Data Lake, Warehouse, Mart

Name	Illustration
Data Lake	
Data Warehouse	
Data Mart	